



(IC1). It starts charging the battery when the battery voltage drops below a preset value. Charging stops when the battery voltage reaches the upper limit. IC TL431, along with a few additional components, can be used as a comparator with hysteresis by taking advantage of its inner voltage reference.

The threshold voltage of the circuit

is the upper limit of charging voltage, which can be calculated as follows:

$$V_T = V_{ref} \times (1 + R1/R2)$$

where V_{ref} , the internal reference voltage of IC TL431, is around 2.5V.

When the battery voltage rises above the threshold voltage, relay RL1 energises to disconnect the charging circuit and blinking LED1 flashes. A hysteresis path is provided

through resistor R3, which controls the relay release voltage (V_R):

$$V_R = V_{ref} \times \frac{1 + [R1 \times R3 / (R1 + R3)]}{R2}$$

When the battery voltage is less than the release voltage, the cathode voltage of TL431 goes to the battery voltage, the relay de-energises, charging continues and LED1 does not flash. LED1 flashes again when the battery voltage after recharging exceeds the

threshold voltage. Resistor R5 limits the battery charging current.

Power supply for the charger is provided by step-down transformer X1 and bridge rectifier comprising diodes D1 through D4.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Use crocodile clips to connect the battery for charging. ●

SKIN RESPONSE METER

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Human skin offers some resistance to current and voltage. This resistance changes with the emotional state of the body. The circuit proposed here measures changes in your skin resistance following changes in your mental state.

In the relaxed state, the resistance offered by the skin is as high as 2 mega-ohms or more, which reduces to 500 kilo-ohms or less when the emotional stress is too high. The reduction in skin resistance is related to increased blood flow and permeability followed by the physiological changes during high stress. This increases the electrical conductivity of the skin.

This circuit is useful to monitor the skin's re-

sponse to relaxation techniques. It is very sensitive and shows response during a sudden moment of stress. Even a deep sigh will give response in the circuit.

The circuit uses a sensitive amplifier to sense variations in the skin resistance. IC CA3140 (IC1) is designed as a resistance-to-voltage converter that outputs varying voltage based on the skin's conductivity. It is wired as an

inverting amplifier to generate constant current to skin in order to measure the skin resistance.

IC CA3140 is a 4.5MHz BiMOS operational amplifier with MOSFET inputs and bipolar output. The gate-protected inputs have high impedance and can sense current as low as 10 pA.

This device is ideal to sense small currents in low-input-current applications.

The inverting input (pin 2) of IC1 is connected to ground (through preset VR1) and one of the touch plates, while the non-inverting input (pin 3) is grounded directly. The output from IC1 passes through current-limiting resistor R1 to the

